

ΠΟΛΥΤΕΧΝΕΙΟ ΚΡΗΤΗΣ
TECHNICAL UNIVERSITY OF CRETE

# REPORT ASSIGNMENT 08

## RANSOMWARE IMPLEMENTATION

Ασφάλεια Συστημάτων και Υπηρεσιών

*Καλανδράνη Όλγα* | *2022030130*

Διδάσκων: Σωτήριος Ιωαννίδης

Σχολή Ηλεκτρολόγων Μηχανικών και Μηχανικών Υπολογιστών, Πολυτεχνείο Κρήτης

ΗΡΥ 413 | Ασφάλεια Συστημάτων και Υπηρεσιών (2025 – 2026)

# Implementation (3.0) :

A dedicated *Docker* image was used to complete this assignment in a controlled and isolated environment. The image is based on Ubuntu and provides all the required dependencies and tools pre-installed, allowing us to complete the assignment without modifying our host system.

```
PS C:\Downloads\ransomware-lab> docker pull kostasamplia/ransom-lab-ubuntu:2025
2025: Pulling from kostasamplia/ransom-lab-ubuntu
7e49dc6156b0: Pull complete
f2b7648b56c1: Pull complete
4f4fb700ef54: Pull complete
06b9700d652c: Pull complete
e23380e50466: Pull complete
de9e741632b6: Download complete
Digest: sha256:3659a42dcc03afe154128c6d879c7b3feb7405cd4b6d829e7cf761340f1eeb51
Status: Downloaded newer image for kostasamplia/ransom-lab-ubuntu:2025
docker.io/kostasamplia/ransom-lab-ubuntu:2025
PS C:\Downloads\ransomware-lab>
```

*Figure 1Docker image download*

Afterwards, on the host system, three local directories were created. The purpose of each directory is the following:

- *./assignment*: Our source code (including the access-logging tool from the prerequisite assignment).
- *./data*: Logs and other generated artifacts that must persist on the host.
- *./target*: Output directory for generated/encrypted test files.

Whilst using the *./target* directory only as a safe directory

```
PS C:\Downloads\ransomware-lab> ls


    Directory: C:\Downloads\ransomware-lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        09/01/2026     12:10                assignment
d-----        09/01/2026     12:10                data
d-----        09/01/2026     12:10                target


PS C:\Downloads\ransomware-lab>
```

*Figure 2The 3 directories*

Lastly, the Docker container was started and the necessary host directories were mounted using bind mounts. This allows the container to access specific folders from the host system, while keeping the rest of the system isolated.

```
PS C:\Downloads\ransomware-lab> docker run --rm -it `
>> -v "${PWD}\assignment:/home/student/assignment" `
>> -v "${PWD}\data:/home/student/data" `
>> -v "${PWD}\target:/home/student/target" `
>> kostasamplia/ransom-lab-ubuntu:2025 `
>> bash
student@6be6bbee7d01:~/assignment$ ls /home/student
assignment  data  target
student@6be6bbee7d01:~/assignment$
```

*Figure 3Container with bind mounts*

## Ransomware Implementation (3.1) :

A bash script named *ransomware.sh* was implemented to simulate a simple ransomware workflow. The script demonstrates the typical stages of a ransomware attack, such as file generation, encryption and deletion of the original plaintext files.

The *ransomware.sh* script was made executable using the *chmod +x* command and executed directly from the cl.

```
student@6be6bbee7d01:~/assignment$ls -l ransomware.sh
-rw-r--r-- 1 student student 905 Jan  9 10:25 ransomware.sh
student@6be6bbee7d01:~/assignment$ chmod +x ransomware.sh
student@6be6bbee7d01:~/assignment$ ls -l ransomware.sh
-rwxr-xr-x 1 student student 905 Jan  9 10:25 ransomware.sh
```

*Figure 4Executable adn executed*

For file encryption, the *OpenSSL* tool was used with the *AES-256-CBC* encryption algorithm. The ransomware script uses the *openssl enc* command with *PBKDF2* to encrypt files.

```
student@6be6bbee7d01:~/assignment$ grep -n "openssl enc" ransomware.sh
48:  openssl enc -aes-256-cbc -salt -pbkdf2 -iter 100000 -in "$f" -out "$out" -k "password123"
student@6be6bbee7d01:~/assignment$ grep -n "pbkdf2" ransomware.sh
48:  openssl enc -aes-256-cbc -salt -pbkdf2 -iter 100000 -in "$f" -out "$out" -k "password123"
student@6be6bbee7d01:~/assignment$
```

*Figure 5Encryption algorithm OpenSSL*

After execution, the script generated the requested files and successfully encrypted them whilst leaving only the encrypted *.enc* files in the target directory.

```
student@6be6bbee7d01:~/assignment$ rm -f /home/student/target/* 2>/dev/null || true
./ransomware.sh /home/student/target 3
[*] Generating 3 files in /home/student/target
[*] Starting encryption in /home/student/target
[+] Encryption completed
student@6be6bbee7d01:~/assignment$ ls -la /home/student/target
total 4
drwxrwxrwx 1 root    root    4096 Jan  9 10:41 .
drwxr-x--- 1 student student 4096 Jan  9 10:16 ..
-rw-r--r-- 1 student student   32 Jan  9 10:41 file_1.txt.enc
-rw-r--r-- 1 student student   32 Jan  9 10:41 file_2.txt.enc
-rw-r--r-- 1 student student   32 Jan  9 10:41 file_3.txt.enc
student@6be6bbee7d01:~/assignment$
```

*Figure 6Successfull Execution*

## Run the Ransomware File in the /data Directory (3.2) :

When the *ransomware.sh* script is executed in the */home/student/data* directory, only the files contained in that directory were affected. The script successfully encrypts the files located in *./data* directory (*a.txt*, *b.txt* files were created for testing), while files in this directory remain untouched. This happens because the ransomware script is restricted to operate only on specific directories, which limits its access and prevents it from modifying other files.

```
student@6be6bbee7d01:~/assignment$ ls -la /home/student/data
total 4
drwxrwxrwx 1 root    root    4096 Jan  9 10:10 .
drwxr-x--- 1 student student 4096 Jan  9 10:16 ..
student@6be6bbee7d01:~/assignment$
```

```
student@6be6bbee7d01:~/assignment$ ./ransomware.sh /home/student/data
[*] Starting encryption in /home/student/data
[+] Encryption completed
student@6be6bbee7d01:~/assignment$ ls -la /home/student/data
total 4
drwxrwxrwx 1 root    root    4096 Jan  9 10:48 .
drwxr-x--- 1 student student 4096 Jan  9 10:16 ..
-rw-r--r-- 1 student student   32 Jan  9 10:48 a.txt.enc
-rw-r--r-- 1 student student   32 Jan  9 10:48 b.txt.enc
student@6be6bbee7d01:~/assignment$
```

*Figure 7Successfull execution*

This demonstrates the principle of least privilege and *role separation*, where restricting user permissions limits the ransomware blast radius. Even if ransomware executes successfully, the impact is limited to the scope of resources accessible by the compromised user.

## Access Control Logging Tool (3.3) :

For access control logging, the shared library *audit_logger.so* from *Assignment 03* was used. The ransomware simulation was executed with *LD_PRELOAD* in order to preload the *audit_logger.so* library.

```
student@6be6bbee7d01:~/assignment$ls -la /home/student/assignment
total 52
drwxrwxrwx 1 root    root     4096 Jan  9 15:00 .
drwxr-x--- 1 student student  4096 Jan  9 10:16 ..
-rwxrwxrwx 1 root    root      432 Nov 10 10:03 Makefile
-rwxrwxrwx 1 root    root     7945 Nov 19 13:13 audit_logger.c
-rwxrwxrwx 1 root    root      507 Nov 19 12:49 audit_logger.h
-rwxrwxrwx 1 root    root    21120 Nov 19 13:36 audit_logger.so
-rwxrwxrwx 1 root    root     6082 Nov 19 13:25 audit_monitor.c
-rwxrwxrwx 1 root    root      628 Nov 19 12:49 audit_monitor.h
-rwxr-xr-x 1 student student  1340 Jan  9 10:38 ransomware.sh
student@6be6bbee7d01:~/assignment$
```

All relevant file operations (create, open, write, close) were recorded in a log file named */tmp/access_audit.log* in append mode.

```
student@6be6bbee7d01:~/assignment$ ls -la audit_logger.so
-rwxr-xr-x 1 student student 21144 Jan  9 15:02 audit_logger.so
```

```
student@6be6bbee7d01:~/assignment$ LD_PRELOAD=/home/student/assignment/audit_logger.so ./ransomware.sh /home/student/tar
get 2
[*] Generating 2 files in /home/student/target
[*] Starting encryption in /home/student/target
[+] Encryption completed
```

```
student@6be6bbee7d01:~/assignment$ find /home/student -name "access_audit.log" -print
student@6be6bbee7d01:~/assignment$ find /home/student -name "access_audit.log" -print
student@6be6bbee7d01:~/assignment$ find / -name "access_audit.log" 2>/dev/null
/tmp/access_audit.log
```

Each log entry includes the user ID (UID), the process ID (PID), the absolute file path, the date and time in UTC, an operation code (0 = create, 1 = open, 2 = write, 3 = close), a denied flag (1 if denied) and a SHA-256 hash of the file contents.

```
student@6be6bbee7d01:~/assignment$ tail -n 20 /tmp/access_audit.log
1000|101|/home/student/target/file_1.txt.enc|2026-01-09|15:05:33|2|0|e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49
5991b7852b855
1000|101|/home/student/target/file_1.txt.enc|2026-01-09|15:05:33|2|0|e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49
5991b7852b855
1000|101|/home/student/target/file_1.txt.enc|2026-01-09|15:05:33|2|0|e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49
5991b7852b855
1000|99|/usr/lib/terminfo/x/xterm|2026-01-09|15:05:33|1|0|07bf300610fcb270364331da44f902162a2329561c251cacd45834984051e0
ca
1000|103|/home/student/target/file_2.txt.enc|2026-01-09|15:05:33|2|0|e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49
5991b7852b855
1000|103|/home/student/target/file_2.txt.enc|2026-01-09|15:05:33|2|0|e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49
5991b7852b855
1000|103|/home/student/target/file_2.txt.enc|2026-01-09|15:05:33|2|0|e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49
5991b7852b855
student@6be6bbee7d01:~/assignment$
```

*Figure 8Output log entry*

## Detect the Ransomware by enriching the Access Control Log Monitoring Tool (3.4) :

The Access Control Monitoring tool *audit_monitor.c* was extended with additional detection logic in order to identify behavioral indicators consistent with ransomware activity. The detection is based on the audit trail produced by the *audit_logger.so* shared library.

Required detection capabilities:

- High Volume File Activity Detection:

  The monitor detects ransomware by checking if many encrypted file ops occur within a short period of time (20 mins). If the number of encrypted files exceeds a given threshold value *X* (provided by input), the activity is reported as suspicious. This reflects typical ransomware behaviour where many files are encrypted rapidly.

- Encryption – and – delete Workflow Detection:

  The monitor identifies ransomware behavior when a file is accessed and an encrypted file *.enc* version of the same file is subsequently created by the same process.

This detection logic is executed by using the *-r <X>* option of the audit monitor (i.e. ./audit_monitor -r 5 ).

```
student@6be6bbee7d01:~/assignment$ gcc -Wall -O2 -o audit_monitor audit_monitor.c -lcrypto
student@6be6bbee7d01:~/assignment$ ./audit_monitor -r 2
=== Ransomware Detection (threshold X=2, window=20min) ===
=== Detection finished ===
student@6be6bbee7d01:~/assignment$ rm -f /home/student/target/* 2>/dev/null || true
LD_PRELOAD=/home/student/assignment/audit_logger.so ./ransomware.sh /home/student/target 10
./audit_monitor -r 5
[*] Generating 10 files in /home/student/target
[*] Starting encryption in /home/student/target
[+] Encryption completed
=== Ransomware Detection (threshold X=5, window=20min) ===
=== Detection finished ===
student@6be6bbee7d01:~/assignment$
```

```
student@6be6bbee7d01:~/assignment$ gcc -Wall -O2 -o audit_monitor audit_monitor.c -lcrypto
student@6be6bbee7d01:~/assignment$ ./audit_monitor -r 2
=== Ransomware Detection (threshold X=2, window=20min) ===
[ALERT] High-volume file creation: 2 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_1.txt.enc
[ALERT] High-volume file creation: 3 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_1.txt.enc
[ALERT] High-volume file creation: 4 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 5 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 6 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 2 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_1.txt.enc
[ALERT] High-volume file creation: 3 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_1.txt.enc
[ALERT] High-volume file creation: 4 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 5 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 6 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 7 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 8 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 9 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 10 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 11 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_3.txt.enc
[ALERT] High-volume file creation: 12 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_3.txt.enc
[ALERT] High-volume file creation: 13 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_3.txt.enc
[ALERT] High-volume file creation: 14 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_4.txt.enc
[ALERT] High-volume file creation: 15 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_4.txt.enc
[ALERT] High-volume file creation: 16 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_4.txt.enc
[ALERT] High-volume file creation: 17 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_5.txt.enc
[ALERT] High-volume file creation: 18 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_5.txt.enc
[ALERT] High-volume file creation: 19 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_5.txt.enc
[ALERT] High-volume file creation: 20 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_6.txt.enc
[ALERT] High-volume file creation: 21 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_6.txt.enc
[ALERT] High-volume file creation: 22 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_6.txt.enc
[ALERT] High-volume file creation: 23 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_7.txt.enc
[ALERT] High-volume file creation: 24 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_7.txt.enc
[ALERT] High-volume file creation: 25 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_7.txt.enc
[ALERT] High-volume file creation: 26 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_8.txt.enc
[ALERT] High-volume file creation: 27 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_8.txt.enc
[ALERT] High-volume file creation: 28 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_8.txt.enc
[ALERT] High-volume file creation: 29 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_9.txt.enc
[ALERT] High-volume file creation: 30 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_9.txt.enc
[ALERT] High-volume file creation: 31 create-events within last 20 minutes (threshold=2)
        Latest created: /home/student/target/file_9.txt.enc
=== Detection finished ===
student@6be6bbee7d01:~/assignment$
```

```
student@6be6bbee7d01:~/assignment$ ls /home/student/target | wc -l
10
student@6be6bbee7d01:~/assignment$ ./audit_monitor -r 5
=== Ransomware Detection (threshold X=5, window=20min) ===
[ALERT] High-volume file creation: 5 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 6 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 5 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 6 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 7 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_10.txt.enc
[ALERT] High-volume file creation: 8 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 9 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 10 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_2.txt.enc
[ALERT] High-volume file creation: 11 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_3.txt.enc
[ALERT] High-volume file creation: 12 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_3.txt.enc
[ALERT] High-volume file creation: 13 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_3.txt.enc
[ALERT] High-volume file creation: 14 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_4.txt.enc
[ALERT] High-volume file creation: 15 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_4.txt.enc
[ALERT] High-volume file creation: 16 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_4.txt.enc
[ALERT] High-volume file creation: 17 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_5.txt.enc
[ALERT] High-volume file creation: 18 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_5.txt.enc
[ALERT] High-volume file creation: 19 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_5.txt.enc
[ALERT] High-volume file creation: 20 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_6.txt.enc
[ALERT] High-volume file creation: 21 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_6.txt.enc
[ALERT] High-volume file creation: 22 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_6.txt.enc
[ALERT] High-volume file creation: 23 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_7.txt.enc
[ALERT] High-volume file creation: 24 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_7.txt.enc
[ALERT] High-volume file creation: 25 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_7.txt.enc
[ALERT] High-volume file creation: 26 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_8.txt.enc
[ALERT] High-volume file creation: 27 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_8.txt.enc
[ALERT] High-volume file creation: 28 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_8.txt.enc
[ALERT] High-volume file creation: 29 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_9.txt.enc
[ALERT] High-volume file creation: 30 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_9.txt.enc
[ALERT] High-volume file creation: 31 create-events within last 20 minutes (threshold=5)
        Latest created: /home/student/target/file_9.txt.enc
=== Detection finished ===
```

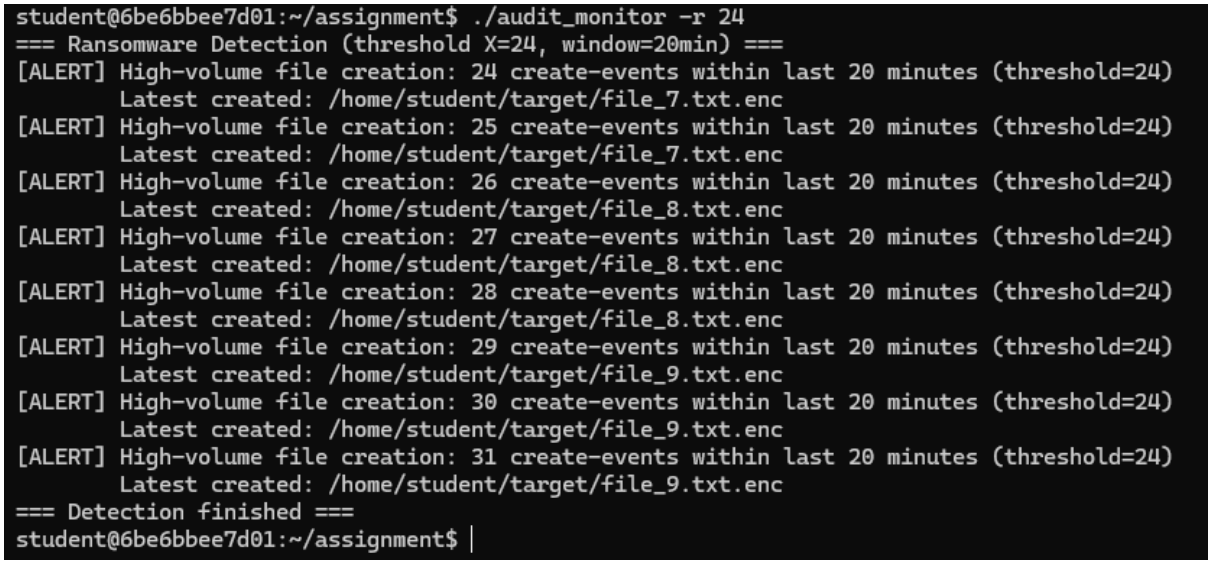



*Figure 9 Outputs with different thresholds*

## Theoretical Questions (3.5) :

1. Why can “least privilege” and role separation reduce ransomware blast radius, even if the initial infection happens on a user workstation? Give a concrete example involving shared drives or administrative credentials.

   Least privilege and role separation reduce ransomware impact by limiting what a compromised user can access. If ransomware infects a user acc, it can only encrypt the files that this user has permission to modify. For example, without access to shared drives or admin rights, ransomware cannot affect system-wide or shared resources.

2. Why do “offline” or “immutable” backups reduce ransomware risk compared to ordinary networkaccessible backups?

   Offline or immutable backups are safer because ransomware cannot access or modify them. Network accessible backups can be encrypted or deleted, while offline backups remain intact and allow data recovery after an attack.

3. Why do many ransomware attacks try to delete backups and recovery points before encrypting files, and what is one simple defense strategy that specifically targets this behavior?

Ransomware attacks, often delete backups to prevent victims from restoring their data without paying. A simple defence strategy, is to use protected backups, such as immutable or offline backups, which ransomware cannot delete.